

5.3 Independence of Path; Conservative Vector Field Worksheet

1. Find $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = (x^2 - y^2)\mathbf{i} - 2xy\mathbf{j}$ and C is the curve $y = x^5 - x^3 + 4x + \sin(2\pi x)$ from $(0, -5)$ to $(1, -1)$.

2. Consider the vector field $\mathbf{F}(x, y) = \left\langle \frac{1}{2}xy, x \right\rangle$.

- a) Compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ where C is the curve $y = 2x$ moving from $(0, 0)$ to $(1, 2)$ in $\mathbb{R}^2$.

- b) Construct two different paths that would connect the points $(0, 0)$ and $(1, 2)$ in $\mathbb{R}^2$.

- c) Determine if $\mathbf{F}$ is independent of path.

3. Determine if each of the following statements is TRUE or FALSE:

a) The union of two simply connected domains is connected.

b) If a differentiable vector field $\mathbf{F}$ satisfies $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path C in an open, connected domain D , then there always exists a function f such that $\nabla f = \mathbf{F}$ on D .

c) If a differentiable vector field $\mathbf{F} = P\mathbf{i} + Q\mathbf{j}$ satisfies $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ on a connected domain D , then $\mathbf{F}$ is conservative on D .

d) If $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for some closed path C , then $\mathbf{F}$ must be a conservative vector field.

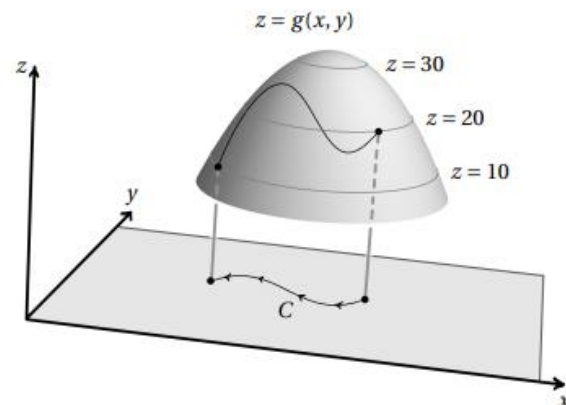
4. Consider the vector field $\mathbf{F}(x, y, z) = \langle e^x + yz, z^2 + xz, 2yz + xy \rangle$.

a) Find a function f on $\mathbb{R}^3$ whose gradient is $\mathbf{F}$.

b) Compute the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is a line segment from $(1, 0, 0)$ to $(0, 0, 1)$.

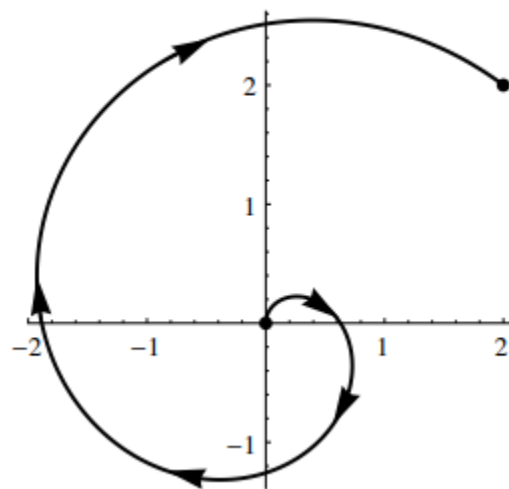
5. Let $\mathbf{F}(x, y) = \langle 2x + y, 3y^2 + e^y \rangle$. Determine whether $\mathbf{F}$ is a conservative vector field. If it is, find its potential function.

6. Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function whose graph is shown at the right, and let C be the indicated curve in the xy -plane. Using only this information evaluate the line integral $\int_C \vec{\nabla} g \cdot d\mathbf{r}$.



7. Consider the vector field $\mathbf{F}(x, y) = \langle y^2, 1 + 2xy \rangle$ on $\mathbb{R}^2$.
- a) Show that $\mathbf{F}$ is conservative by finding its potential function $f(x, y)$.

- b) For the curve C shown at the right, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$.



8. Consider the curve C which is a straight-line segment from $(0,0)$ to $(1,1)$.

a) For $\mathbf{F} = \langle x, -y^2 \rangle$, compute $\int_C \mathbf{F} \cdot d\mathbf{r}$.

b) For $f(x,y) = \cos\left(\frac{\pi}{2}xy\right)e^{xy}$, compute $\int_C \vec{\nabla} f \cdot d\mathbf{r}$.

c) Circle the best estimate for $\frac{1}{\sqrt{2}} \int_C e^{xy/200} \cos\left(\frac{xy}{200}\right) ds$.

-3	-2	-1	0	1	2	3
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9. Let $\mathbf{F}(x, y)$ be the vector field defined by $\mathbf{F}(x, y) = \begin{cases} \langle 0, 1 \rangle & \text{if } y > 0 \\ \langle 0, 0 \rangle & \text{if } y = 0 \\ \langle 0, -1 \rangle & \text{if } y < 0 \end{cases}$.

- a) Let C be the unit circle oriented counterclockwise. Show that $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$. (You will need to split the integral into two pieces).

- b) It turns out that $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ is true for any closed curve C in the plane and not just the unit circle. But, this vector field is *not* conservative. Explain why this vector field cannot be conservative.

10. Let D be the set of all point (x, y) in $\mathbb{R}^2$ except for $(0, 0)$. In each part below, indicate whether a continuous vector field with domain D and the property described is *necessarily conservative* or *not necessarily conservative*. (If the vector field is never conservative, circle *not necessarily conservative*.)

a) $\mathbf{F}_1(x, y) = \langle -y, x \rangle$.

$\mathbf{F}_1$ is

necessarily
conservative

not necessarily
conservative

b) $\mathbf{F}_2$ has the path independence property; that is, the line integral $\int_C \mathbf{F}_2 \cdot d\mathbf{r}$ is independent in D .

$\mathbf{F}_2$ is

necessarily
conservative

not necessarily
conservative

c) $\mathbf{F}_3 = \langle P, Q \rangle$ where $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$ over D .

$\mathbf{F}_3$ is

necessarily
conservative

not necessarily
conservative

d) $\mathbf{F}_4$ has the property that for the unit circle C_1 (given by $x^2 + y^2 = 1$), $\int_{C_1} \mathbf{F}_4 \cdot d\mathbf{r} = 0$.

$\mathbf{F}_4$ is

necessarily
conservative

not necessarily
conservative

11. A continuous vector field $\mathbf{F}$, defined on $\mathbb{R}^2$, is shown at the right. For the sake of understanding the scale in this picture let $\mathbf{F}(0,0) = \langle 0, 0.1 \rangle$. Assuming that $\mathbf{F}$ is conservative, circle the value of $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is the curve shown from $(0,-1)$ to $(0,1)$.

-0.3	-0.2	-0.1	0	0.1	0.2	0.3
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